📝 DAV Lab Assignment 3: Seaborn Visualizations

Subject: Data Analysis & Visualization (DAV) | **Dataset:** MoviesOnStreamingPlatforms.csv | **Tool:** Python (Pandas, Seaborn, Matplotlib)

> 📌 **Lab Objective & Transition:**
> Previously completed on dummy data (`earthquake.csv`) plotting *Alert Level vs Magnitude*.
> Now adapted directly for **`MoviesOnStreamingPlatforms.csv`** comparing *Age Rating vs Rotten Tomatoes Scores & platform correlations!*

🔍 Mapping Old Variables ➔ New Dataset

- **Categorical Feature:** **`Age`** (*Rating:* `all`, `7+`, `13+`, `16+`, `18+`) *[Replaces 'alert']*
- **Numerical Feature:** **`Rotten Tomatoes`** *score* *[Replaces 'magnitude']*
- **Correlation Matrix:** `['Year', 'Rotten Tomatoes', 'Netflix', 'Hulu', 'Prime Video', 'Disney+']`

⚡ Step 0: Setup, Libraries & Essential Data Cleaning

In the CSV, Rotten Tomatoes is saved as text like `"98/100"`. We strip `"/100"` and convert it into a numeric `float` so Seaborn can compute quartiles!

```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Load dataset
df = pd.read_csv("MoviesOnStreamingPlatforms.csv")

# 2. Clean Rotten Tomatoes column: "98/100" → 98.0
df['Rotten Tomatoes'] = df['Rotten Tomatoes'].astype(str).str.replace('/100', '')
df['Rotten Tomatoes'] = pd.to_numeric(df['Rotten Tomatoes'], errors='coerce')

# 3. Drop missing values in Age and Rotten Tomatoes
plot_df = df.dropna(subset=['Age', 'Rotten Tomatoes']).copy()
plot_df = plot_df[plot_df['Age'].str.strip() ≠ '']

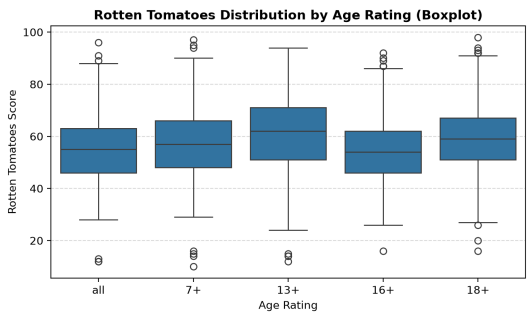
# 4. Set chronological maturity order
age_order = ['all', '7+', '13+', '16+', '18+']
```

> 💡 **Why define `age_order`?** *Prevents alphabetical sorting (which puts 13+ before 7+) and ensures logical maturity progression!*

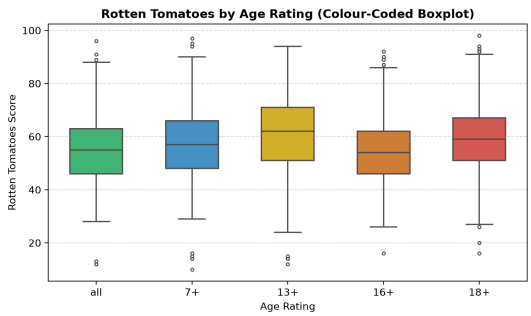
📦 Step 1 & 1b: Boxplot & Colour-Coded Boxplot

What is a Boxplot? *Displays the 5-number summary:* **Minimum, Q1 (25%), Median (Q2 50%), Q3 (75%), and Maximum**. *Points past 1.5×IQR are plotted as outliers.*

Standard Boxplot



Color-Coded Boxplot



💻 Implementation Code:

```
palette = {
    'all': '#2ecc71',   # Green
    '7+':  '#3498db',   # Blue
    '13+': '#f1c40f',   # Yellow
    '16+': '#e67e22',   # Orange
    '18+': '#e74c3c'    # Red
}

plt.figure(figsize=(10, 6))
sns.boxplot(
    data=plot_df,
    x='Age',
    y='Rotten Tomatoes',
    order=age_order,
    palette=palette,
    width=0.6,
    linewidth=1.2,
    fliersize=3
)
plt.title('Rotten Tomatoes Distribution by Age Rating (Colour-Coded Boxplot)')
plt.xlabel('Age Rating')
plt.ylabel('Rotten Tomatoes Score')
plt.tight_layout()
plt.show()
```

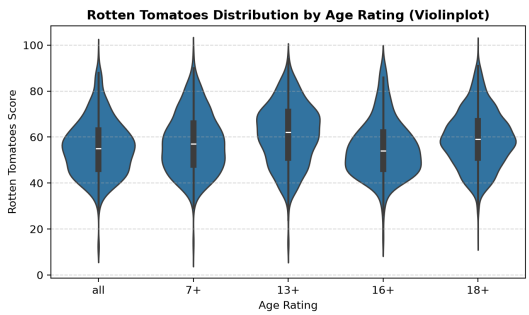
🧠 **Anatomy & Observations:**

- **Median:** *The horizontal line inside each box sits around ~48–52 across all Age groups.*
- **Fliers/Dots:** *Movies with scores above 90 appear as upper outliers (critical masterpieces).*

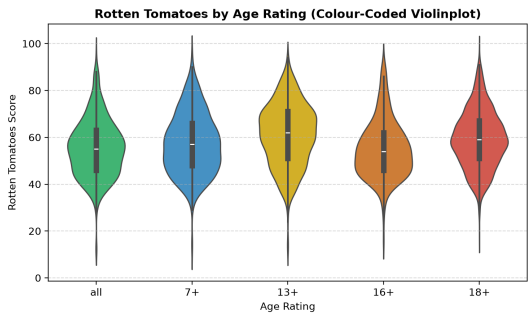
🎻 Step 2 & 2b: Violinplot & Colour-Coded Violinplot

Why use a Violin Plot? A violin plot combines the statistical box plot with a rotated **Kernel Density Estimate (KDE)** on each side. The thickness shows where values are clustered!

Standard Violinplot



Color-Coded Violinplot



💻 Implementation Code:

```
plt.figure(figsize=(10, 6))
sns.violinplot(
    data=plot_df,
    x='Age',
    y='Rotten Tomatoes',
    order=age_order,
    palette=palette,
    width=0.6,
    linewidth=1.2
)
plt.title('Rotten Tomatoes by Age Rating (Colour-Coded Violinplot)')
plt.xlabel('Age Rating')
plt.ylabel('Rotten Tomatoes Score')
plt.tight_layout()
plt.show()
```

🔍 **Graph Insights:** Notice the pronounced middle bulge in each category between 40 and 55, proving most movies receive moderate ratings with a thin tail towards 90+.

Step 3: Histplot with KDE Curve

Analyzes the univariate continuous distribution of `Rotten Tomatoes` *scores across the entire streaming catalog.*

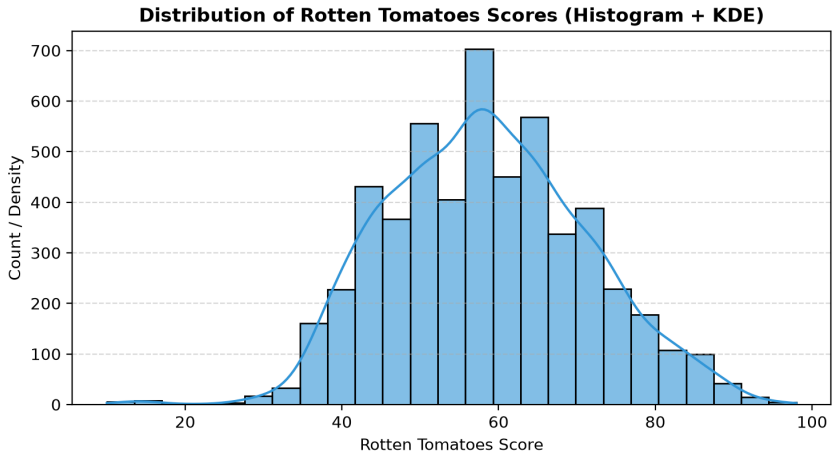


Figure: Histogram (bins=25) + Kernel Density Estimation (KDE) curve

```
plt.figure(figsize=(8, 5))
sns.histplot(
    data=plot_df,
    x='Rotten Tomatoes',
    kde=True,
    bins=25,
    color='#3498db'
)
plt.title('Distribution of Rotten Tomatoes Scores (Histogram + KDE)')
plt.xlabel('Rotten Tomatoes Score')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```

🎯 **What `kde=True` does:** *Draws a smooth probability density curve over the discrete bins, making it easy to spot skewness or multimodality.*

🔥 Step 4: Correlation Matrix & HeatMap

Calculates Pearson pairwise correlation coefficients between numeric variables (-1 = inverse, 0 = none, +1 = direct).

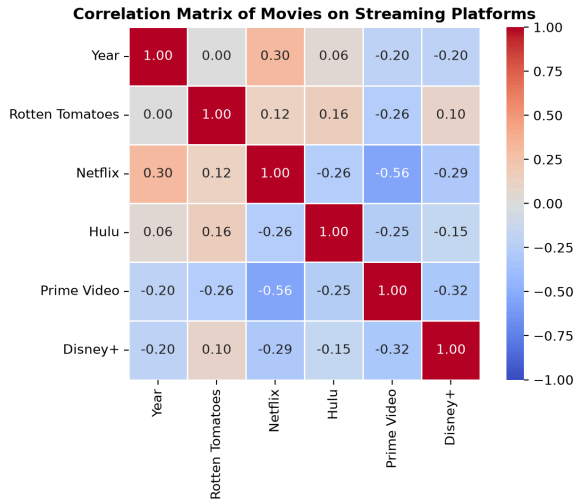


Figure: Correlation Heatmap with annot=True and coolwarm diverging colormap

```
nums_cols = ['Year', 'Rotten Tomatoes', 'Netflix', 'Hulu', 'Prime Video', 'Disney+']
corr = plot_df[nums_cols].corr()

plt.figure(figsize=(7, 5.5))
sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm',
    vmin=-1,
    vmax=1,
    fmt=".2f",
    square=True
)
plt.title('Correlation Matrix (Year, Rotten Tomatoes, Streaming Platforms)')
plt.tight_layout()
plt.show()
```

🔑 ***Parameters:*** `annot=True` *displays correlation numbers,* `vmin=-1, vmax=1` *normalizes scale, and* `fmt=".2f"` *limits decimals.*

🎓 Complete Lab Viva Questions & Answers (Part 1)

Q1. What is the fundamental difference between Matplotlib and Seaborn?

Ans: Matplotlib is a low-level graphics library that provides fine-grained control over every pixel and line, requiring verbose code. Seaborn is built directly on top of Matplotlib and pandas; it specializes in statistical visualization, automatic statistical aggregations, built-in themes, and color palettes in just one or two lines.

Q2. What is the Five-Number Summary in a Boxplot?

Ans: It consists of: (1) **Minimum** (Q1 - 1.5×IQR), (2) **First Quartile (Q1 / 25th percentile)**, (3) **Median (Q2 / 50th percentile)**, (4) **Third Quartile (Q3 / 75th percentile)**, and (5) **Maximum** (Q3 + 1.5×IQR).

Q3. How are Outliers mathematically identified in a Boxplot?

Ans: By using the **IQR (Interquartile Range)** rule: `IQR = Q3 - Q1`.
Any data point less than `Q1 - 1.5 × IQR` or greater than `Q3 + 1.5 × IQR` is treated as an outlier and plotted as a distinct point (flier).

Q4. When would a Boxplot fail where a Violinplot succeeds?

Ans: In **bimodal or multimodal distributions**! A boxplot only shows the median and quartiles, so a bimodal distribution (two peaks) can look identical to a uniform distribution in a boxplot. A violin plot shows the actual probability density curve (KDE), making two separate peaks immediately visible.

Q5. What is Kernel Density Estimation (KDE)?

Ans: KDE is a non-parametric way to estimate the probability density function (PDF) of a continuous random variable. It places a smooth kernel (typically Gaussian) over each data point and sums them up to create a continuous, smooth curve over discrete histogram bins.

🎓 Complete Lab Viva Questions & Answers (Part 2)

Q6. Why did we have to clean the 'Rotten Tomatoes' column?

Ans: The raw column contained strings like `"98/100"` (object data type). Statistical functions like calculating means, medians, quartiles, and Pearson correlation require numeric data (float or int). If uncleaned, Seaborn throws a `TypeError` or treats values as discrete categorical text.

Q7. What is Pearson Correlation and what do values of -1, 0, and +1 mean?

Ans: It measures the linear relationship between two continuous variables.
- **+1:** Perfect positive linear correlation (as X increases, Y increases).
- **0:** No linear correlation.
- **-1:** Perfect negative linear correlation (as X increases, Y decreases).

Q8. If correlation between two features is 0, does it mean they are completely independent?

Ans: No! Pearson correlation only detects *linear* relationships. Two variables can have a strong non-linear relationship (e.g. $y = x^2$) and still yield a correlation score close to 0.

Q9. Explain the key parameters used in `sns.heatmap()`:

Ans:
- `annot=True`: Writes numerical correlation values inside each cell.
- `fmt=".2f"`: Formats numbers as floats with 2 decimal places.
- `cmap='coolwarm'`: Diverging palette (cool blue = negative, warm red = positive).
- `vmin=-1, vmax=1`: Fixes the color bounds to the theoretical limits of correlation.
- `square=True`: Forces each cell to be a square aspect ratio.

Q10. Why are streaming platform correlations slightly negative in the heatmap?

Ans: The columns (Netflix, Hulu, Prime Video, Disney+) are binary (1 = available, 0 = not). The negative correlation indicates platform exclusivity—if a movie is licensed on Netflix (1), it is rarely simultaneously licensed on Hulu or Disney+ (0).

⭐ **Examiner Pro-Tip:** Always mention that *"Correlation does not imply causation"* and that *"Violin plots are more informative than box plots because they reveal distribution shape (multimodality) rather than just summary statistics."*